Question 1

a

N	D(b)	D(d)	D(g)	D(f)	D(a)	D(c)	D(h)
e	8	2	2	6	∞	∞	∞
e,d	4	2	2	5	∞	∞	∞
edg	4	2	2	5	∞	8	3
edgh	4	2	2	4	∞	8	3
edghb	4	2	2	4	7	5	3
edghbf	4	2	2	4	7	5	3
edghbfc	4	2	2	4	6	5	3
edghbfca	4	2	2	4	6	5	3
Pre Node	d	e	e	h	c	b	g

b Distance Vector

	D(b)	D(d)	D(g)	D(f)	D(a)	D(c)	D(h)
	8	2	∞	6	∞	∞	∞
	4|D	2|D	∞	5|D	11|B	9|B	7|F
3rd	4|D	2|D	7|F	5|D	7|D	5|D	6|F
4th	4|D	2|D	7|F	5|D	6|D	5|D	6|F

c Poisoned Reverse

D(b)	D(d)	D(g)	D(f)	D(a)	D(c)	D(h)
8	2	∞	6	∞	∞	∞
4|D	2|D	∞	5|D	11|B	9|B	7|F
4|D	2|D	7|F	5|D	7|D	5|D	6|F
4|D	2|D	7|F	5|D	6|D	5|D	6|F

CamScanner

d Cost of Message

~~Co~~ → Costs of path to sourse

→ Cost of source to multicast group

Costs of path to source

from (e) to (b)

= 4

Cost to (c) from (b)

= 1

Cost to (g) from (b)

= 6

Cost to (h) from (b)

= 6

Total Cost = 4 + 1 + 6 + 6

= 17

2 B = link rate × burst time

$= 10^9$ bps × 0.5×10^{-3} s

$= 5 \times 10^5$ bits

Question 3

Entry 2

10. 23. 192. 0

11111111 11111111 11111100 00000000

255 255 . 252 . 0

11111111 11111111 11111100 00000000

00001110 00010111 11000000 00000000

(a) 10. 23. 196. 128

does not match any other entry but Entry 1

Forward to next hop router 10.52.13.42

Interface B

(b) Not in range for Entry 4 and 3

So forward using entry 2

Forward using destination host's link layer address

Interface A

c 10.23.194.63.
This address is inside entry 4 address range
Forward to next hop router 10.23.192.1
Interface A

d 10.23.193.128
It is ~~At~~ Entry 2 ~~on~~ address range
Directly connected network
Forward using destination host's link layer address
Interface A

CamScanner